HotRC F-06A

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HotRC F-06A 6-Channel 2.4GHz FHSS Receiver Instruction Manual

Model: F-06A

Introduction

This manual provides detailed instructions for the HotRC F-06A 6-channel 2.4GHz FHSS receiver. The F-06A is designed for use with various HotRC transmitters, including CT-6A, DS600, and HT-6A, and is suitable for RC cars, boats, aircraft, and tanks. It features a voltage return function, allowing for real-time monitoring of the power battery voltage and receiver signal.

Key Features

- **6-Channel Operation:** Provides 6 independent channels for controlling various functions of your RC model.
- **2.4GHz FHSS Technology:** Ensures reliable and interference-free communication with your transmitter.
- **Voltage Return Function:** The F-06A receiver is equipped with a voltage return function, allowing you to monitor the power battery voltage and receiver signal strength. This requires connecting the voltage return interface and cable.
- **Extended Range:** Offers a receiving distance of approximately 300 meters on the ground and up to 800 meters in the air, though actual performance may vary with environmental conditions.
- **Compact and Lightweight:** Designed for easy integration into various RC models.

Specifications (F-06A)

Attribute	Value
Brand Name	HotRC
Product Model	F-06A
Number of Channels	6
Wireless Frequency	2.4GHz
Input Voltage	DC 4V-9V
Working Current	35mA
Return Voltage Range	0-27V (effective up to 300m)
Return Information	Signal / Voltage (effective up to 300m)
Remote Control Distance	Ground: 300m, Air: 800m (approximate, varies by environment)
Gyroscope	No
Appearance Size	38mm * 24mm * 13mm
Product Weight	7.4g
Applicable Types	Car, Ship, Aircraft, Tank

Setup and Installation

1. Component Identification

Familiarize yourself with the receiver's ports and buttons:

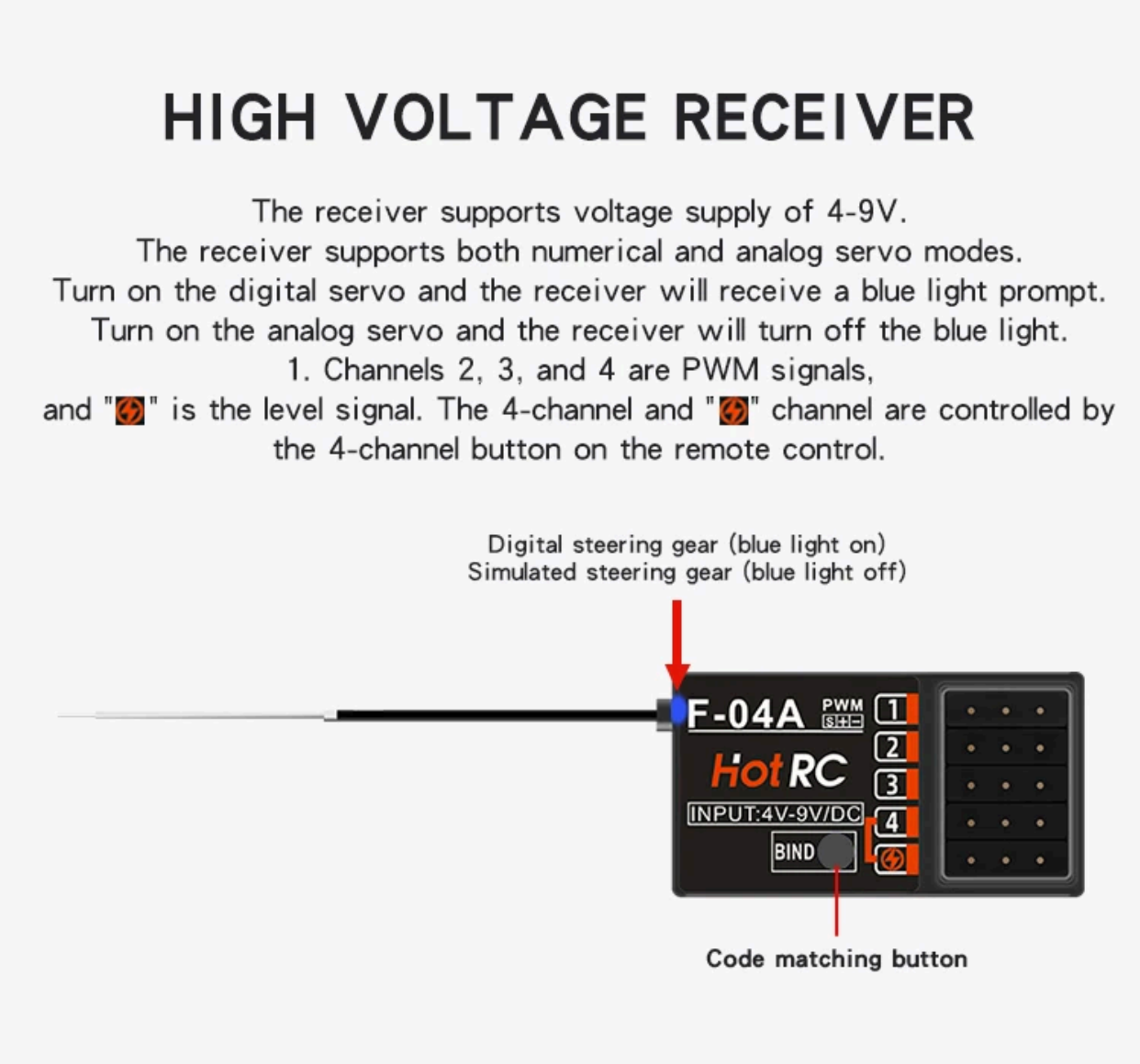


Image: HotRC F-06A receiver showing PWM channels 1-6, EXT-V port, and BIND button.

- **PWM Channels (1-6):** Standard servo/ESC connections.
- **EXT-V:** External Voltage input for telemetry.
- **BIND Button:** Used for pairing the receiver with the transmitter.
- **Antenna:** For signal reception. Ensure it is positioned correctly and not obstructed.

2. Wiring Diagram

The following diagram illustrates typical connections for the F-06A receiver, including power, servos, and the voltage return interface.

Image: Detailed wiring diagram showing connections for the F-06A receiver, including power input (DC 4-9V), servo connections (Digital Servo: blue light bright, Analog Servo: blue light extinct), voltage return interface, and the code matching button.

- **Power Input:** Connect a DC 4V-9V power source to the designated input pins (S, +, -). The diagram shows the positive (+) and negative (-) terminals.
- **Servo Connections:** Connect your digital or analog servos to the PWM channels (1-6).
    - Digital Servos: Indicated by a bright blue light.
    - Analog Servos: Indicated by an extinct (off) blue light.
- **Voltage Return Interface:** For the voltage return function, connect the provided voltage return cable to the EXT-V port on the receiver and to your model's power battery. This allows the transmitter to display the battery voltage.
- **Code Button (BIND):** This button is used for the binding process.

3. Binding Procedure (Pairing Receiver with Transmitter)

To establish communication between your HotRC transmitter and the F-06A receiver, follow these steps:

1. Ensure your transmitter is powered off.

2. Connect power to the F-06A receiver (DC 4V-9V).
3. Press and hold the **BIND** button on the F-06A receiver.
4. While holding the BIND button, power on the receiver. The receiver's LED indicator should start flashing rapidly, indicating it's in binding mode.
5. Release the BIND button.
6. Power on your HotRC transmitter (e.g., CT-6A, DS600, HT-6A) and put it into binding mode according to your transmitter's specific instructions.
7. Once the binding is successful, the receiver's LED indicator will turn solid, and your transmitter should show a successful connection.
8. Test all channels to ensure proper control before operating your model.

Operating Instructions

1. After successful binding and installation, ensure all connections are secure.
2. Power on your transmitter first, then power on your RC model (which powers the receiver).
3. Verify that all control surfaces and functions respond correctly to transmitter inputs.
4. If using the voltage return function, monitor the battery voltage displayed on your transmitter to ensure safe operation.
5. Always perform a range check before operating your model at a distance.
6. Power off your RC model first, then power off your transmitter after use.

Maintenance

- **Keep Dry:** Protect the receiver from moisture and water, as it is not waterproof.
- **Avoid Extreme Temperatures:** Do not expose the receiver to excessively high or low temperatures.
- **Cleanliness:** Keep the receiver free from dust, dirt, and debris. Use a soft, dry brush or cloth for cleaning.
- **Inspect Connections:** Regularly check all wiring and connectors for signs of wear, corrosion, or loose connections.
- **Antenna Care:** Ensure the antenna is not bent sharply, cut, or damaged. Position it away from metal parts for optimal signal reception.

Troubleshooting

Problem	Possible Cause	Solution
Receiver LED not lighting up	No power or incorrect polarity.	Check power connections and ensure correct voltage (DC 4V-9V).
No control response from model	Receiver not bound to transmitter, or signal interference.	Perform the binding procedure again. Ensure transmitter and receiver are within range and away from strong interference sources.
Short control range	Damaged antenna, interference, or improper antenna placement.	Inspect receiver antenna for damage. Reposition antenna away from carbon fiber or metal parts. Avoid operating near Wi-Fi routers or other 2.4GHz devices.
Incorrect servo movement	Incorrect servo connection or reversed channel on	Verify servo connections to the correct PWM channels. Check transmitter settings for channel reverse.

Problem	Possible Cause	Solution
	transmitter.	
Voltage return not working	Voltage return cable not connected or faulty.	Ensure the voltage return cable is securely connected to the EXT-V port and the power battery. Check the cable for damage.

Warranty and Support

HotRC products are manufactured to high quality standards. In case of any issues, please refer to the troubleshooting section first. For further assistance or warranty claims, please contact your retailer or the official HotRC support channels. Keep your purchase receipt as proof of purchase.

Please note that damage caused by misuse, unauthorized modification, or improper installation is not covered under warranty.

Related Documents - F-06A

	[HotRC CT-6A 2.4GHz RC Transmitter: Features, Specs, and Guide] Comprehensive overview of the HotRC CT-6A 6-channel RC transmitter, detailing its features, power options, specifications, and configuration guide for model cars, ships, and tanks.
	[HotRC DS600 6-Channel Remote Control Quick Start Guide] A comprehensive quick start guide for the HotRC DS600 6-channel remote control system, covering setup, operation, and specifications for various receiver models.
	[HotRC DS600 Quick Start Guide - 6 Channel Radio Set] A quick start guide for the HotRC DS600, a 6-channel 2.4GHz radio set for model boats and fishing boats. Includes setup, operation, and specifications.

Hotrc Hot rc HT-10A HT10A 2.4Ghz 10CH Remote Controller Transmitter Mode 2（left hand）

Specifications:

Screen: 1.7-inch color screen
Applicable Models: Model airplanes/cars/boats, etc.
Channels: 10 channels
Modulation Mode: GFSK
Battery: 25 7.4V Lipo4.1 5AA (not included)
RC Distance: Approx. 360m on the ground800m in the air
Transmit Power: ≤ 120MA
Net Weight: 320g
Spread Spectrum Mode: FHSS
Response Speed: Pulse Width Modulation: <20ms
Frequency Range: 2.4GHz ISM
Receiver Sensitivity: -96 dBm
Receiver Power Supply: DC 3.0V-8V
Transmitter Power Supply: DC 3.7V-9V

Features:

1. 1.7-inch color screen, 10CH remote control receiver with standard return function.

[Hotrc HT-10A 2.4GHz 10CH Remote Controller Transmitter - Specifications & Features]

Detailed specifications and features of the Hotrc HT-10A 2.4GHz 10-channel remote controller transmitter, suitable for model airplanes, cars, boats, drones, and more. Includes information on screen, range, power, and functionality.

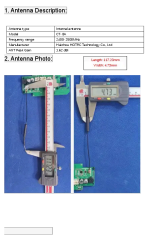
Hotrc F-10A receiver

Specification:

Product model: F-10A
Number of channels: 10
Wireless frequency: 2.4GHz
Working current: 35mA
Receiving distance: 300-800m* Feedback information: Signal/Voltage【300m】
input voltage:DC4V-9V
size:36*23*13mm
weight:8.5g
Return voltage: 0-27V【300m】
gyroscope: no
Applicable types : Car/ship/plane/tank

[Hotrc F-10A Receiver Specifications - 2.4GHz 10-Channel RC Receiver]

Detailed technical specifications for the Hotrc F-10A 2.4GHz 10-channel receiver, including its wireless frequency, working current, receiving distance, input voltage, dimensions, weight, and compatible vehicle types for RC cars, ships, planes, and tanks.



1. Antenna Description:

Antenna type	Internal antenna
Gain	2.0dBi
Frequency range	2400-2500MHz
Manufacturer	Huizhou HOTRC Technology Co., Ltd
APP Field Use	LAN IR

2. Antenna Photo:

Length: 47.20mm
Width: 4.75mm

[CT-8A Internal Antenna Specification and Test Report]

Detailed specifications and test results for the CT-8A internal antenna, manufactured by Huizhou HOTRC Technology Co., Ltd. Covers frequency range, peak gain, and radiation patterns.